MC Quantum Trajectory Benchmarks for Cavity-OAT

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1 Introduction

We consider a driven-dissipative cavity coupled to an ensemble of three-level atoms for generating spin squeezing through a Berry phase that depends on the active atom number. We use an effective spin model obtained by adiabatically eliminating the cavity in the bad-cavity limit. Each atom has two long-lived states, $|\uparrow\rangle$ and $|\downarrow\rangle$, and an excited state $|e\rangle$. The drive couples $|\downarrow\rangle \leftrightarrow |e\rangle$, while $|\uparrow\rangle$ is uncoupled. Therefore, for a fixed number of atoms in the active manifold,

$$N_J = N_\downarrow + N_e, \tag{1}$$

the dynamics reduce to an effective two-level system in the $\{|\downarrow\rangle, |e\rangle\}$ manifold.

Due to the resulting strong symmetry, a pure state in the case of homogeneous atom-drive coupling may be decomposed into subspaces with a fixed number of atoms in the active manifold, N_J, as

$$|\psi(t)\rangle = \sum_{N_J=0}^{N} c_{N_J}(t)\ |\psi_{N_J}(t)\rangle\,, \qquad \sum_{N_J=0}^{N} |c_{N_J}(t)|^2 = 1. \tag{2}$$

The coherent dynamics within each N_J subspace are governed, in the rotating frame of the drive, by the active-manifold Hamiltonian

$$\hat{H}_\delta = |\Omega|\hat{J}_x - \delta\hat{N}_e, \tag{3}$$

where Ω is the effective Rabi frequency, δ is the detuning of the drive, and the excited-state number operator $\hat{N}_e$ and collective spin operator $\hat{J}_x$ are defined as

$$\hat{N}_e = \sum_{i=1}^{N_J} |e\rangle\langle e|_i\,, \qquad \hat{J}_x = \frac{1}{2}\sum_{i=1}^{N_J} \left(|\downarrow\rangle\langle e|_i + |e\rangle\langle\downarrow|_i\right). \tag{4}$$

In the case of two distinct atom-drive coupling strengths, specified by the Hamiltonian

$$\hat{H}_\delta = |\Omega|\left(\omega_1\hat{J}_{1,x} + \omega_2\hat{J}_{2,x}\right) - \delta\hat{N}_e, \tag{5}$$

where $\omega_1 + \omega_2 = 2$, a pure state may similarly be decomposed into subspaces with fixed numbers of atoms in the active manifold for each coupling strength, $N_J = N_{J_1} + N_{J_2}$, as

$$|\psi(t)\rangle = \sum_{N_J=0}^{N}\sum_{N_{J_1}=0}^{N_J} c_{N_J,N_{J_1}}(t)\ |\psi_{N_J,N_{J_1}}(t)\rangle\,, \qquad \sum_{N_J=0}^{N}\sum_{N_{J_1}=0}^{N_J} |c_{N_J,N_{J_1}}(t)|^2 = 1. \tag{6}$$

The three-stage protocol is implemented as

$$\begin{cases} \text{Phase 1:} & \Omega(t) = \Omega_0, & \delta(t) = 0, \\ \text{Phase 2:} & \Omega(t) = \Omega_0, & \delta(t) = \delta_0, \\ \text{Phase 3:} & \Omega(t) = 0, & \delta(t) = 0. \end{cases} \tag{7}$$

Phase 1 prepares the dissipative steady state in the active manifold. Phase 2 applies a detuning and accumulates a sector-dependent Berry phase. Phase 3 turns off the drive and allows the excited population to decay back to $|\downarrow\rangle$.